



FACULTÉ DES SCIENCES ET DES TECHNOLOGIES(FST)

TD N°8_ – RÉSEAUX 2

COURS: RÉSEAUX 2

PRÉNOM: Lens Sandro

NOM: PÉTIOTE

NIVEAU: L3

DATE: 17/05/2026

L'objectif de ce TD est de:

Installer pfSense sur VirtualBox

Configurer les interfaces réseau (WAN / LAN)

Accéder à l'interface Web (GUI)

Configurer DHCP et NAT

Créer des règles de firewall

Tester la connectivité réseau

Comprendre et appliquer les règles avancées du pare-feu de PfSense.

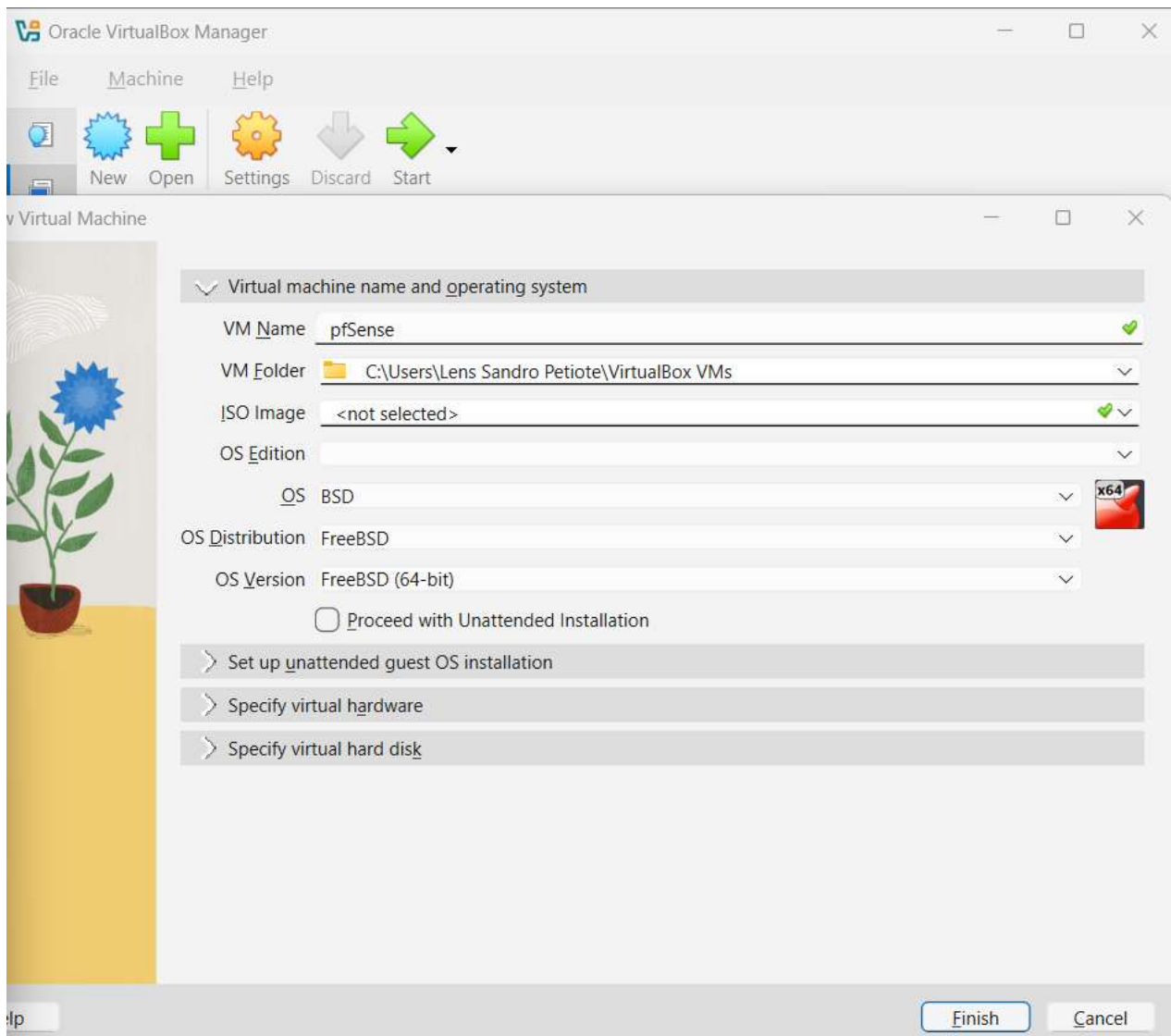
Bloquer ou autoriser des flux spécifiques (ports, IP, protocoles).

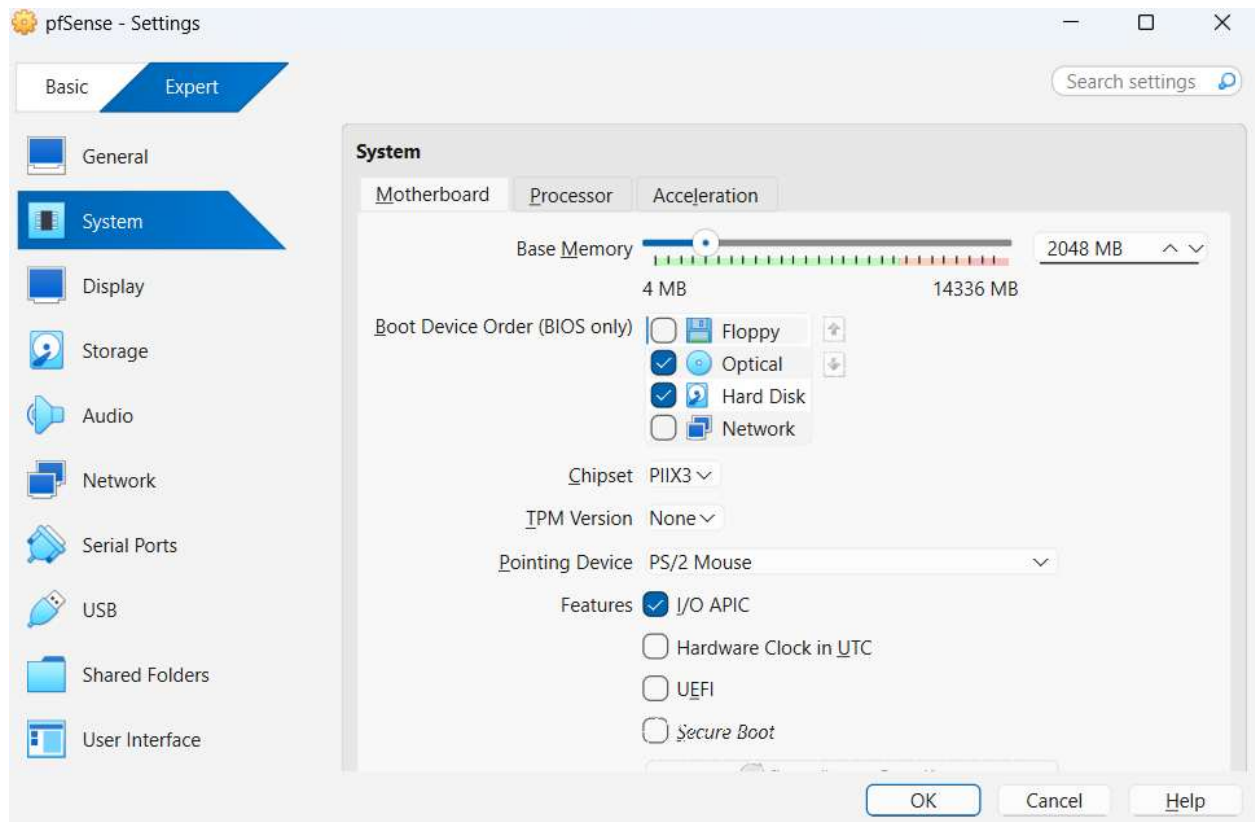
Créer et configurer des horaires de contrôle réseau (Schedules).

Appliquer des règles temporelles via le firewall.

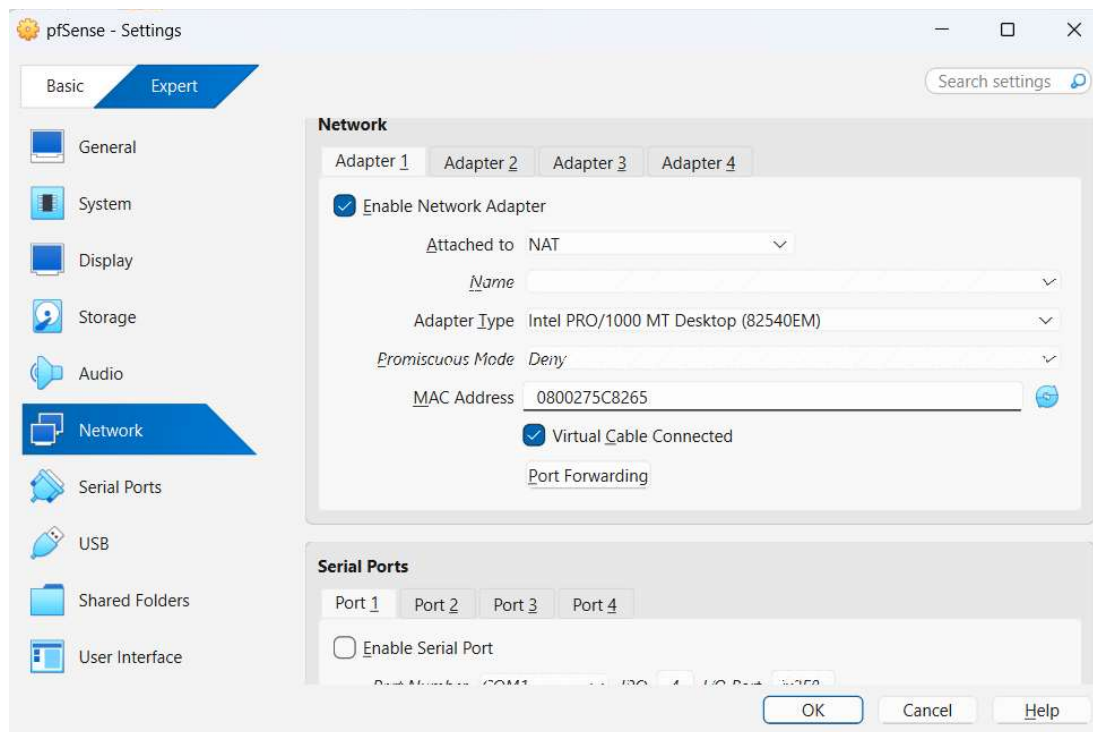
Gérer l'accès à Internet selon des plages horaires (ex : usage bureau/école).

INSTALLATION DE PFSense SUR VIRTUALBOX





Configuration des interfaces réseau



pfSense - Settings

Basic

Expert

Search settings

General

System

Display

Storage

Audio

Network

Serial Ports

USB

Shared Folders

User Interface

network

Adapter 1

Adapter 2

Adapter 3

Adapter 4

☒ Enable Network Adapter

Attached to

Internal Network

Name

LAN_NET

Adapter Type

Intel PRO/1000 MT Desktop (82540EM)

Promiscuous Mode

Deny

MAC Address

08002713525A

☒ Virtual Cable Connected

Serial Ports

Port 1

Port 2

Port 3

Port 4

☐ Enable Serial Port

Port Number

COM1

I/O

4

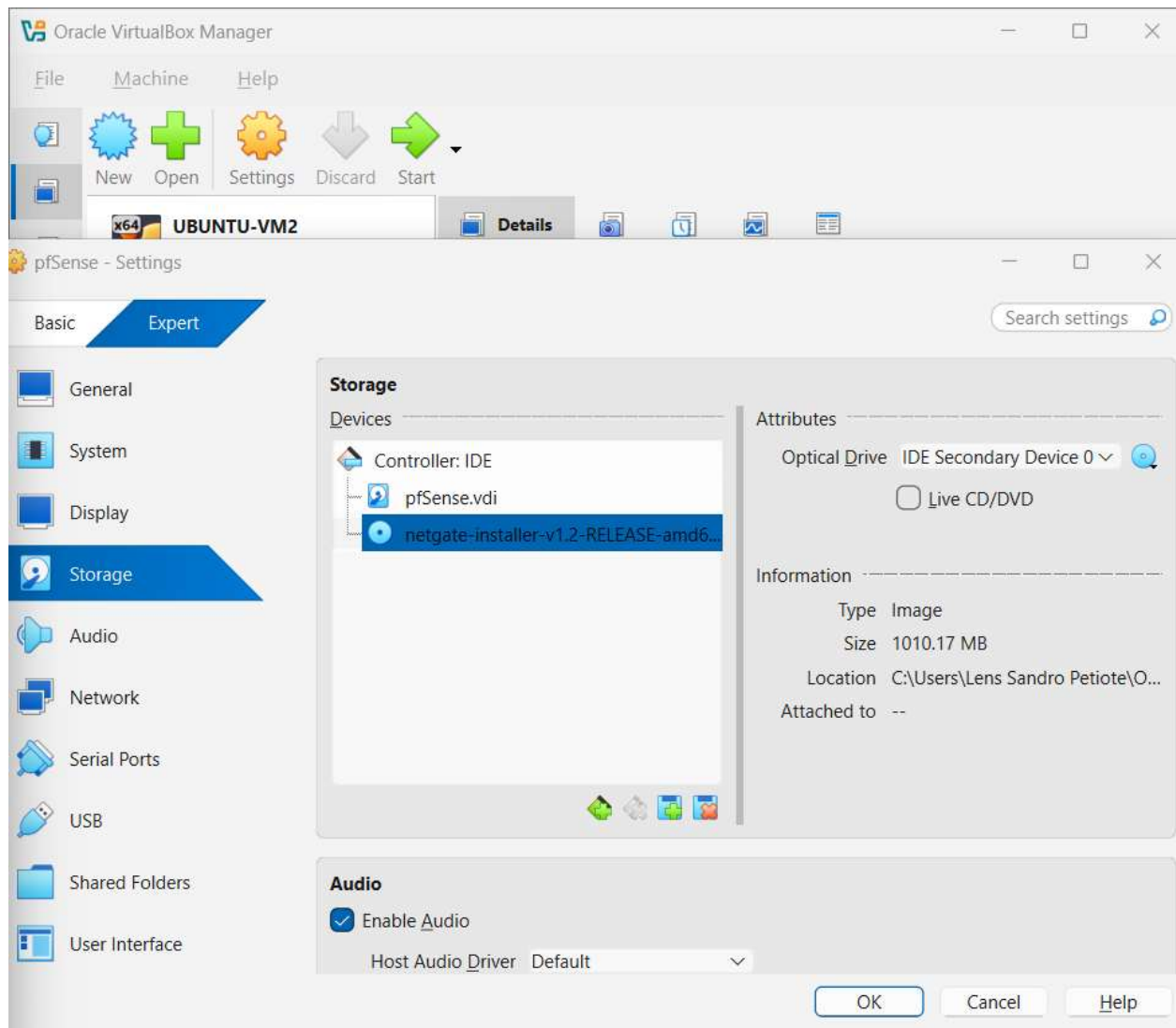
I/O Port

ix3F8

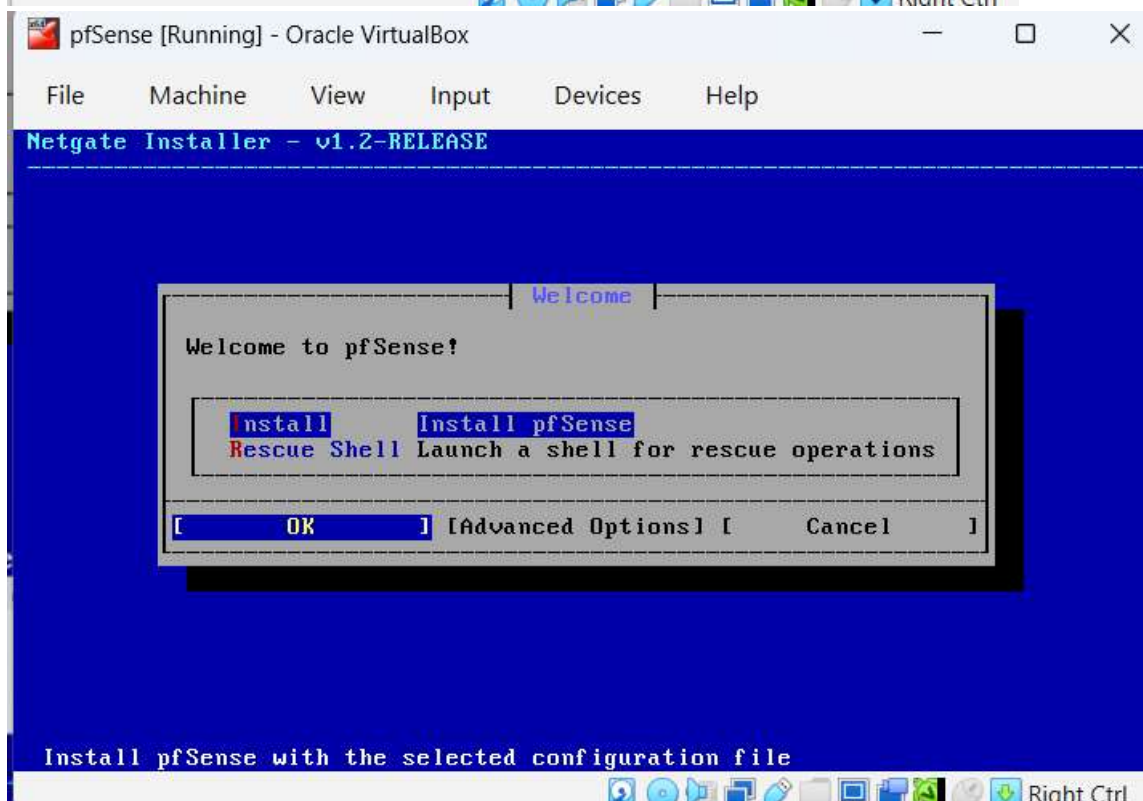
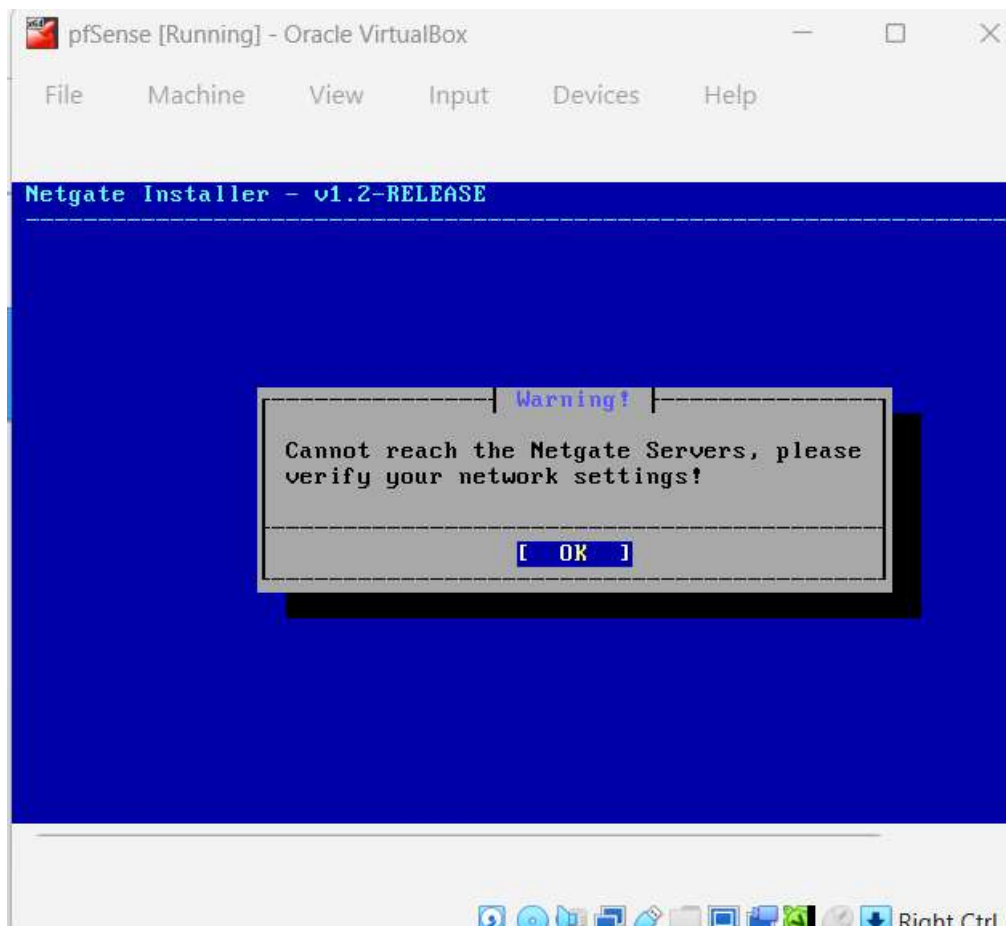
OK

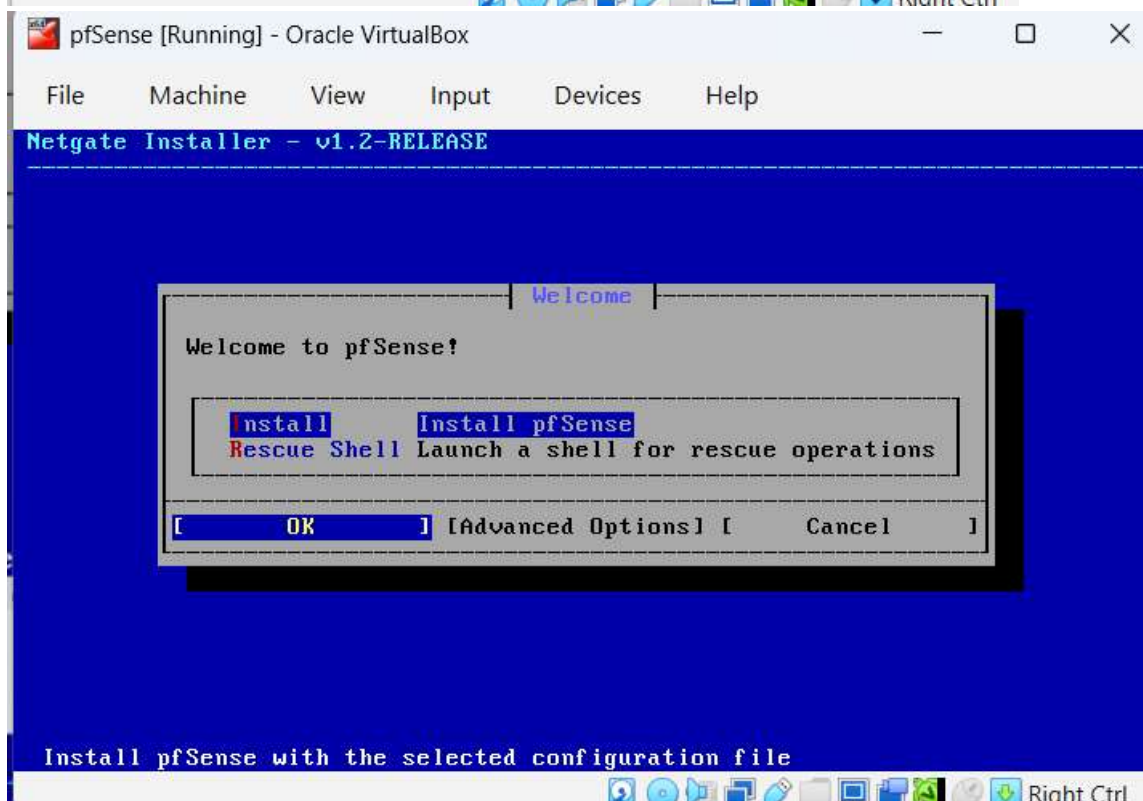
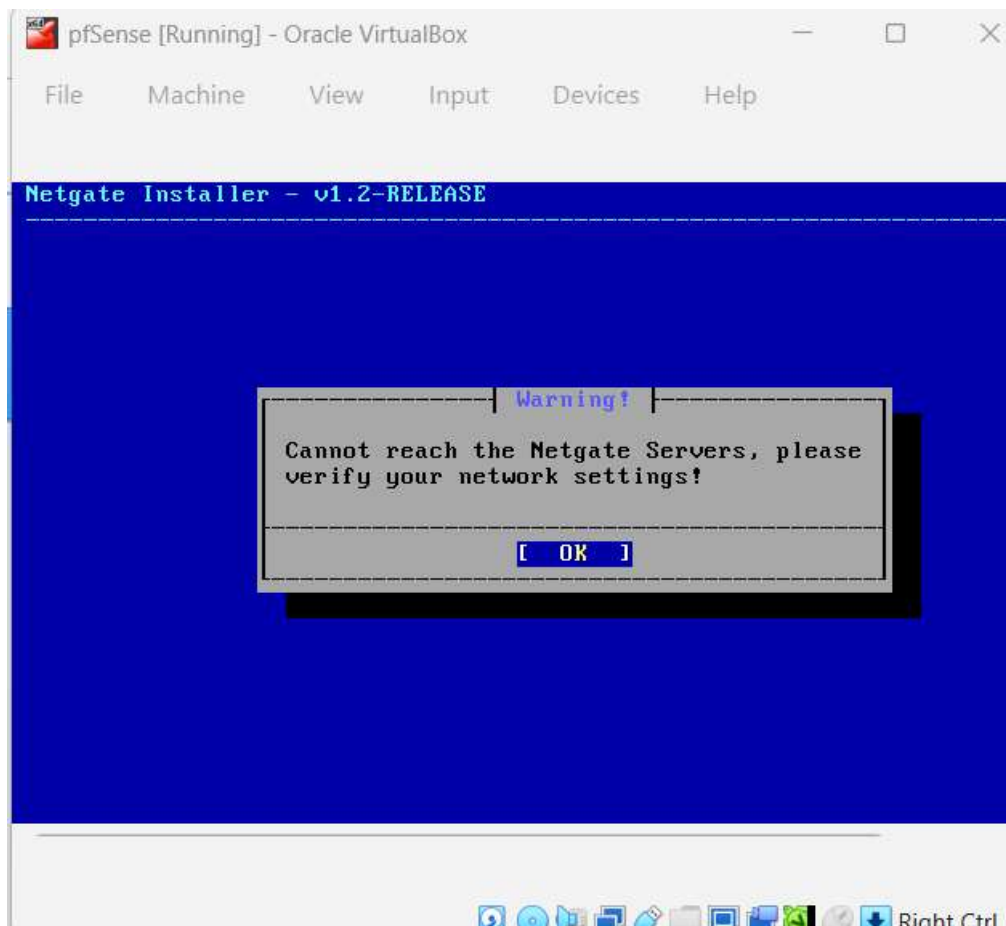
Cancel

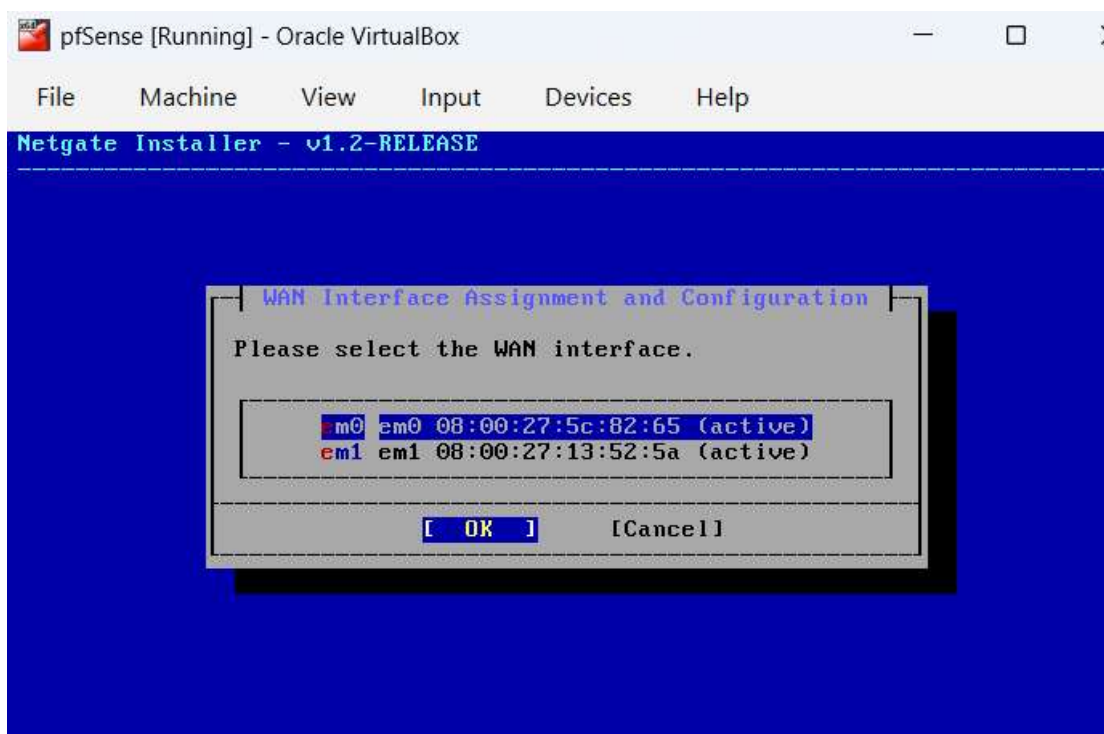
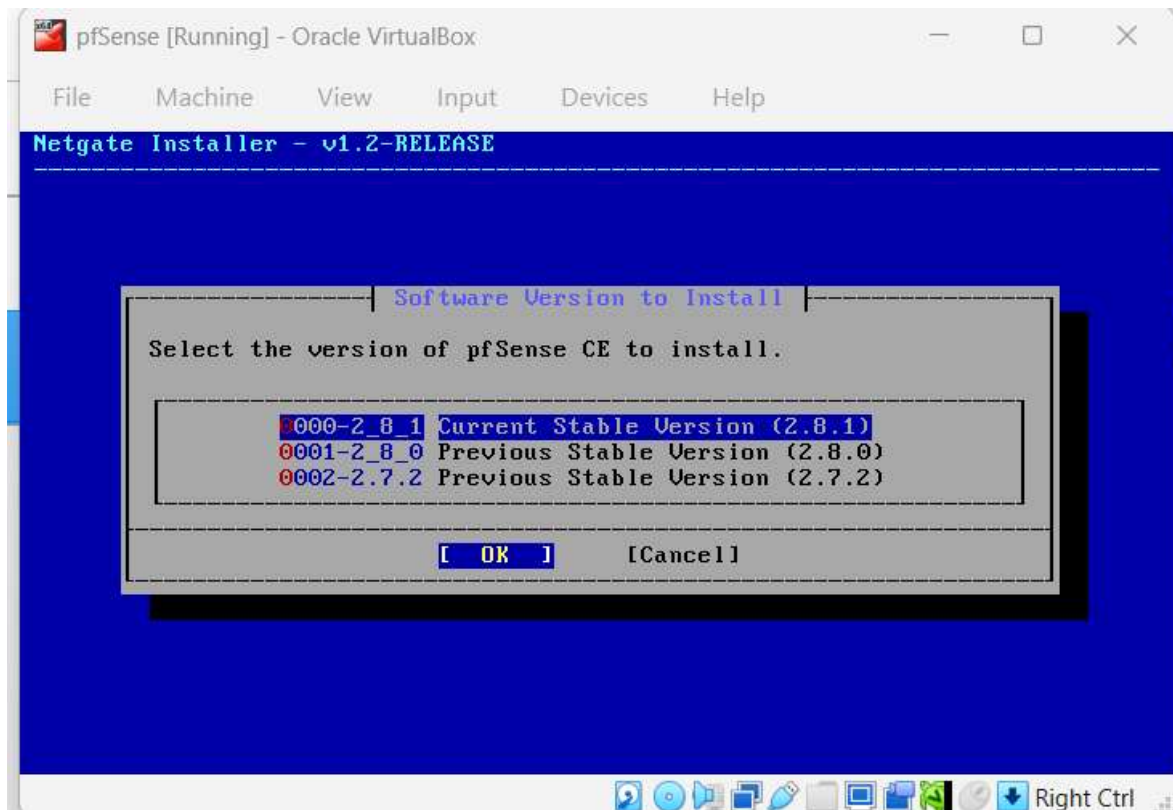
Help



INSTALLATION DE PFSENSE







```
pfSense [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Starting CRON... done.
pfSense 2.8.1-RELEASE amd64 20260513-1943
Bootup complete

FreeBSD/amd64 (pfSense.home.arpa) (ttyv0)

VirtualBox Virtual Machine - Netgate Device ID: 2186e06fcedc6d9f6da5

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

WAN (wan) -> em0 -> v4/DHCP4: 10.0.2.15/24
v6/DHCP6: fd17:625c:f037:2:a00:27ff:fe5c:8265/64
LAN (lan) -> em1 -> v4: 192.168.1.1/24

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address      11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults        13) Update from console
5) Reboot system                   14) Enable Secure Shell (sshd)
6) Halt system                     15) Restore recent configuration
7) Ping host                       16) Restart PHP-FPM
8) Shell

Enter an option: 
```

```
The IPv4 LAN address has been set to 192.168.2.1/24

The IPv6 LAN address has been set to dhcp6

Press <ENTER> to continue.
VirtualBox Virtual Machine - Netgate Device ID: 2186e06fcedc6d9f6da5

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

WAN (wan) -> em0 -> v4/DHCP4: 10.0.2.15/24
v6/DHCP6: fd17:625c:f037:2:a00:27ff:fe5c:8265/64
LAN (lan) -> em1 -> v4: 192.168.2.1/24

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address      11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults        13) Update from console
5) Reboot system                   14) Enable Secure Shell (sshd)
6) Halt system                     15) Restore recent configuration
7) Ping host                       16) Restart PHP-FPM
8) Shell

Enter an option: 
```

ACCÈS INTERFACE WEB

Configuration dans fedora comme machine cliente

Cancel **Wired** Apply

Details Identity **IPv4** IPv6 Security

IPv4 Method

☐ Automatic (DHCP)
 ☐ Link-Local Only
 ☒ Manual
 ☐ Disable
 ☐ Shared to other computers

Addresses

Address	Netmask	Gateway
192.168.2.10	255.255.255.0	192.168.2.1

DNS Automatic ☒

8.8.8.8

Separate IP addresses with commas

Routes Automatic ☒

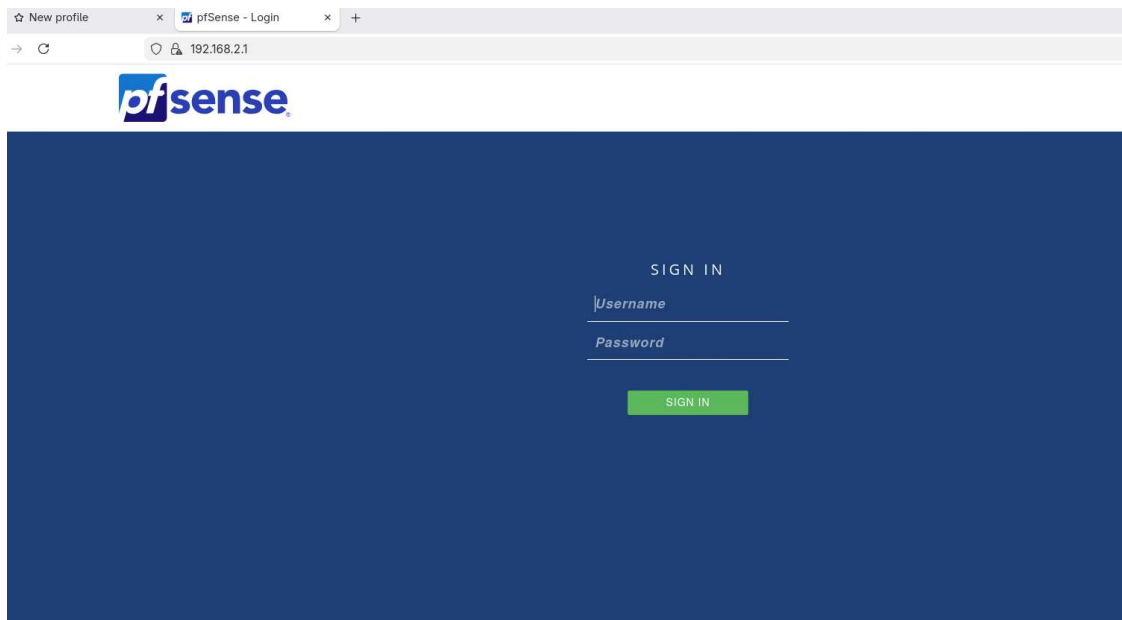
Address	Netmask	Gateway	Metric

Ping sur mon serveur pour tester la connectivité

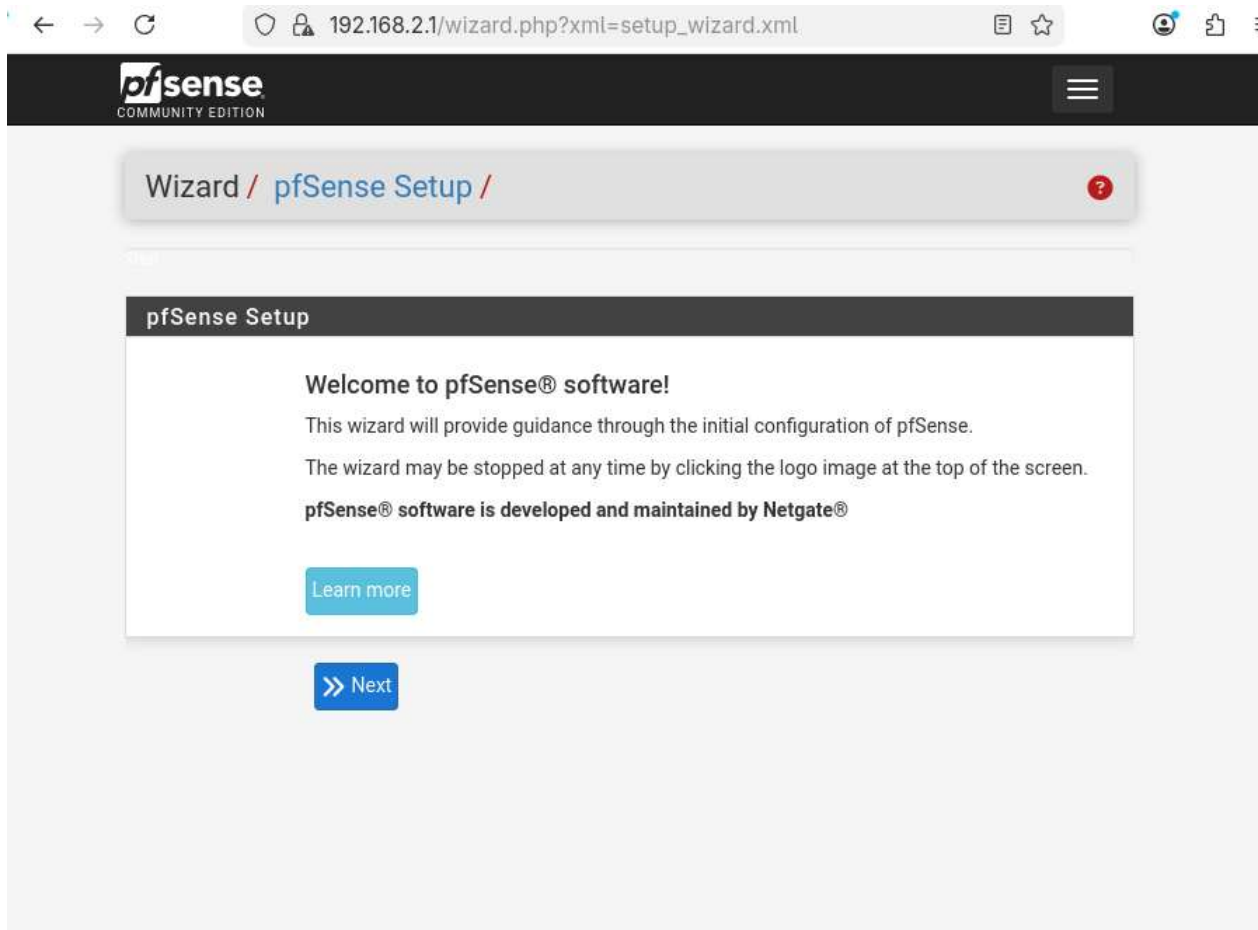
```

May 19  4:14 PM
projet_systemes@fedora:~$ ping 192.168.2.1
PING 192.168.2.1 (192.168.2.1) 56(84) bytes of data.
64 bytes from 192.168.2.1: icmp_seq=1 ttl=64 time=3.31 ms
64 bytes from 192.168.2.1: icmp_seq=2 ttl=64 time=2.23 ms
64 bytes from 192.168.2.1: icmp_seq=3 ttl=64 time=2.83 ms
64 bytes from 192.168.2.1: icmp_seq=4 ttl=64 time=1.42 ms
64 bytes from 192.168.2.1: icmp_seq=5 ttl=64 time=1.85 ms
64 bytes from 192.168.2.1: icmp_seq=6 ttl=64 time=1.06 ms
64 bytes from 192.168.2.1: icmp_seq=7 ttl=64 time=1.52 ms
64 bytes from 192.168.2.1: icmp_seq=8 ttl=64 time=1.27 ms
^Z
[1]+  Stopped                  ping 192.168.2.1
projet_systemes@fedora:~$
  
```

Connexion au compte pour avoir accès à l'interface web



Configuration



192.168.2.1/wizard.php?xml=setup_wizard.xml

On this screen the general pfSense parameters will be set.

Hostname

pfSense

Name of the firewall host, without domain part.

Examples: pfsense, firewall, edgefw

Domain

home.arpa

Domain name for the firewall.

Examples: home.arpa, example.com

Do not end the domain name with '.local' as the final part (Top Level Domain, TLD). The 'local' TLD is widely used by mDNS (e.g. Avahi, Bonjour, Rendezvous, Airprint, Airplay) and some Windows systems and networked devices. These will not network correctly if the router uses 'local' as its TLD. Alternatives such as 'home.arpa', 'local.lan', or 'mylocal' are safe.

The default behavior of the DNS Resolver will ignore manually configured DNS servers for client queries and query root DNS servers directly. To use the manually configured DNS servers below for client queries, visit Services > DNS Resolver and enable DNS Query Forwarding after completing the wizard.

Primary DNS Server

8.8.8.8

Secondary DNS Server

Override DNS

☒

Allow DNS servers to be overridden by DHCP/PPP on WAN

192.168.2.1/wizard.php?XMI=Setup_Wizard.XMI

Address

pptplocalsubnet

32

PPTP Remote IP Address

PPTP Dial on demand

☐ Enable Dial-On-Demand mode

This option causes the interface to operate in dial-on-demand mode, allowing a virtual full time connection. The interface is configured, but the actual connection of the link is delayed until qualifying outgoing traffic is detected.

PPTP Idle timeout

If no qualifying outgoing packets are transmitted for the specified number of seconds, the connection is brought down. An idle timeout of zero disables this feature.

RFC1918 Networks

Block RFC1918 Private Networks

☒ Block private networks from entering via WAN

When set, this option blocks traffic from IP addresses that are reserved for private networks as per RFC 1918 (10/8, 172.16/12, 192.168/16) as well as loopback addresses (127/8). This option should generally be left turned on, unless the WAN network lies in such a private address space, too.

Block bogon networks

Block bogon networks

☒ Block non-Internet routed networks from entering via WAN

When set, this option blocks traffic from IP addresses that are reserved (but not RFC 1918) or not yet assigned by IANA. Bogons are prefixes that should never appear in the Internet routing table, and obviously should not appear as the source address in any packets received.

>> Next

192.168.2.1/wizard.php?xml=setup_wizard.xml

pfSense
COMMUNITY EDITION

Wizard / pfSense Setup / Configure LAN Interface

Step 5 of 9

Configure LAN Interface

On this screen the Local Area Network information will be configured.

LAN IP Address	192.168.2.1
Type dhcp if this interface uses DHCP to obtain its IP address.	
Subnet Mask	24

>> Next

192.168.2.1/wizard.php?xml=setup_wizard.xml

pfSense
COMMUNITY EDITION

Wizard / pfSense Setup / Reload configuration

Step 7 of 9

Reload configuration

Click 'Reload' to reload pfSense with new changes.

>> Reload

Wizard completed.**Congratulations! pfSense is now configured.**

We recommend that you check to see if there are any software updates available. Keeping your software up to date is one of the most important things you can do to maintain the security of your network.

[Check for updates](#)**Remember, we're here to help.**

[Click here](#) to learn about Netgate 24/7/365 support services.

User survey

Please help all the people involved in improving and expanding pfSense software by taking a moment to answer this short survey (all answers are anonymous)

[Anonymous User Survey](#)

Useful resources.

- Learn more about Netgate's product line, services, and pfSense software from our [website](#)
- To learn about Netgate appliances and other offers, [visit our store](#)
- Become part of the pfSense community. Visit our [forum](#)
- Subscribe to our [newsletter](#) for ongoing product information, software announcements and special offers.

[Finish](#)

new profile x pfSense.nome.arpa - St x +

→ 192.168.2.1/index.php ☆

pfSense
COMMUNITY EDITION

Status / Dashboard + ?

Disks ⚙️ - ×

Mount	Used	Size	Usage
> /	898M	12G	7% of 12G (zfs)

Interfaces ⚙️ - ×

WAN	↑	1000baseT <full-duplex>	10.0.2.15 fd17:625c:f037:2:a00:27ff:fe5c:8265
LAN	↑	1000baseT <full-duplex>	192.168.2.1

CONFIGURATION FIREWALL

Création du Schedule

new profile

pfSense.home.arpa - Fir

192.168.2.1/firewall_schedule_edit.php

Month

May_26

Date

May_2026						
Mon	Tue	Wed	Thu	Fri	Sat	Sun
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

Click individual date to select that date only. Click the appropriate weekday Header to select all occurrences of that weekday.

Time

0

00

23

59

Start HrsStart MinsStop HrsStop Mins

Select the time range for the day(s) selected on the Month(s) above. A full day is 0:00-23:59.

Time range description

A description may be entered here for administrative reference (not parsed).

+ Add Time

Clear selection

Configured Ranges

Mon - Fri	8:00	18:00		Delete
Day(s)	Start time	Stop time	Description	

Save

Browser tabs: New profile, pfSense.home.arpa - Firefox

Address bar: 192.168.2.1/firewall_schedule.php

pfSense COMMUNITY EDITION

Firewall / Schedules

Schedules			
Name	Range: Date / Times / Name	Description	Actions
 WORKING_HOURS	Mon - Fri / 8:00-18:00 /	Acces Internet autorise pendant les heures de travail	 

 Indicates that the schedule is currently active.

 Add



Utilisation du Schedule dans une règle firewall

Edit Firewall Rule

Action

Block

Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled

☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

LAN

Choose the interface from which packets must come to match this rule.

Address

IPv4

Family

Select the Internet Protocol version this rule applies to.

Protocol

TCP

Choose which IP protocol this rule should match.

Source

Source

☐ Invert match

LAN subnets

Source Address /

Display Advanced

The **Source Port Range** for a connection is typically random and almost never equal to the destination port. In most cases this setting must remain at its default value, **any**.

Sync	This does NOT prevent the rule from being overwritten on Slave.	
VLAN Prio	none Choose 802.1p priority to match on.	
VLAN Prio Set	none Choose 802.1p priority to apply.	
Schedule	WORKING_HOURS Leave as 'none' to leave the rule enabled all the time.	
Gateway	Default Leave as 'default' to use the system routing table. Or choose a gateway to utilize policy based routing. Gateway selection is not valid for "IPV4+IPV6" address family.	
In / Out pipe	none	none
	Choose the Out queue/Virtual interface only if In is also selected. The Out selection is applied to traffic leaving the interface where the rule is created, the In selection is applied to traffic	

Bloquer l'accès à YouTube par IP sur pfSense

New profile

pfSense.home.arpa - Firefox

192.168.2.1/firewall_rules_edit.php?if=wan&after=-1

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

Any

Choose which IP protocol this rule should match.

Source

Source

☐ Invert match

Any

Source Address /

Destination

Destination

☒ Invert match

Address or Alias

142.250.0.0

Extra Options

Log

☐ Log packets that are handled by this rule

Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the [Status: System Logs: Settings](#) page).

Description

A description may be entered here for administrative reference. A maximum of 52 characters will be used in the ruleset label and displayed in the firewall log.

Advanced Options

Display Advanced

Save

Créer un Alias

pfSense COMMUNITY EDITION

Firewall / Aliases / Edit

Properties

Name
The name of the alias may only consist of the characters "a-z, A-Z, 0-9 and _".

Description
A description may be entered here for administrative reference (not parsed).

Type

Host(s)

Hint Enter as many hosts as desired. Hosts must be specified by their IP address or fully qualified domain name (FQDN). FQDN hostnames are periodically re-resolved and updated. If multiple IPs are returned by a DNS query, all are used. An IP range such as 192.168.1.1-192.168.1.10 or a small subnet such as 192.168.1.16/28 may also be entered and a list of individual IP addresses will be generated.

IP or FQDN

Test

```
[2]~$ ping 142.250.0.1
projet_systemes@fedora:~$ ping 142.250.0.0
PING 142.250.0.0 (142.250.0.0) 56(84) bytes of data.
From 10.0.2.2 icmp_seq=206 Destination Net Unreachable
From 10.0.2.2 icmp_seq=207 Destination Net Unreachable
From 10.0.2.2 icmp_seq=208 Destination Net Unreachable
From 10.0.2.2 icmp_seq=209 Destination Net Unreachable
From 10.0.2.2 icmp_seq=210 Destination Net Unreachable
From 10.0.2.2 icmp_seq=211 Destination Net Unreachable
From 10.0.2.2 icmp_seq=212 Destination Net Unreachable
From 10.0.2.2 icmp_seq=213 Destination Net Unreachable
From 10.0.2.2 icmp_seq=214 Destination Net Unreachable
From 10.0.2.2 icmp_seq=215 Destination Net Unreachable
From 10.0.2.2 icmp_seq=216 Destination Net Unreachable
```

On remarque que le ping sur Youtube echoue

Pfsense
COMMUNITY EDITION

Firewall / Rules / Edit

Edit Firewall Rule

Action Block
Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled ☐ Disable this rule
Set this option to disable this rule without removing it from the list.

Interface LAN
Choose the interface from which packets must come to match this rule.

Address Family IPv4
Select the Internet Protocol version this rule applies to.

Protocol Any
Choose which IP protocol this rule should match.

Source

Source ☒ Invert match LAN subnets Source Address /

Destination

Destination ☒ Invert match Address or Alias Destination Address /

Right Ctrl

Questions

1. Quel est le rôle de pfSense ?

Le rôle de pfSense est de servir de pare-feu et de routeur, en sécurisant le réseau, en gérant le trafic et en permettant la configuration de règles de sécurité et de NAT.

2. Différence WAN et LAN ?

Le WAN correspond au réseau externe, généralement Internet, tandis que le LAN est le réseau interne local d'une organisation ou d'un domicile. Le LAN relie les machines entre elles, et le WAN connecte ce réseau au monde extérieur.

3. Pourquoi utiliser NAT automatique ?

Le NAT automatique est utilisé pour traduire les adresses IP privées du LAN en une adresse publique unique afin d'accéder à Internet. Cela simplifie la configuration et protège les machines internes en masquant leurs adresses réelles.

4. Comment fonctionne une règle firewall ?

Une règle de firewall fonctionne en filtrant le trafic réseau selon des critères comme l'adresse IP, le port ou le protocole. Elle décide si un paquet est autorisé ou bloqué, garantissant ainsi la sécurité du réseau.

5. Pourquoi VirtualBox est utilisé pour les tests ?

VirtualBox est utilisé pour les tests car il permet de créer des environnements virtuels isolés, où l'on peut expérimenter des configurations réseau sans affecter le système réel. C'est un outil pratique pour l'apprentissage et la simulation.

6. Quel est le rôle de pfSense dans ce réseau ?

Dans ce réseau, pfSense joue le rôle de pare-feu principal et de routeur. Il gère la connexion entre le LAN et le WAN, applique les règles de sécurité, effectue le NAT et assure la protection globale du trafic.

7. Pourquoi utiliser un Schedule

Un Schedule dans pfSense permet de définir des plages horaires pendant lesquelles une règle firewall est active. Cela sert à automatiser le contrôle du trafic réseau : par exemple, autoriser l'accès Internet uniquement pendant les heures de bureau et le bloquer en dehors de ces heures. En résumé, un schedule est utilisé pour appliquer des règles de sécurité de manière temporaire et planifiée.

8. Différence entre PASS et BLOCK dans les logs

- **PASS** signifie que le trafic a été autorisé par une règle firewall. Dans les logs, cela indique que la connexion est passée avec succès.
- **BLOCK** signifie que le trafic a été refusé par une règle firewall. Dans les logs, cela montre que la tentative de connexion a été stoppée.

Coclusion.

Grâce à ce TD j'ai pu Installer pfSense sur VirtualBox, configurer les interfaces réseau (WAN / LAN), accéder à l'interface Web (GUI), configurer DHCP et NAT, appliquer des règles temporelles via le firewall et gérer l'accès à Internet selon des plages horaires. Donc la tâche a été réussie.